MIL-D-5813B(USAF)
13 JULY 1987
SUPERSEDING
MIL-D-5813A(USAF)
3 May 1965

MILITARY SPECIFICATION

DOME, NAVIGATOR'S OBSERVING, NON-PRESSURIZED AIRCRAFT, TYPE A-3A

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of navigator's observing dome, designated Type A-3A.

2. APPLICABLE DOCUMENTS.

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-5425	Plastic, Sheet Acrylic, Heat Resistant
MIL-P-6997	Plastic, Working and Installation of Transparent Sheet, General Specification

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio, ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8576	Adhesive, Acrylic Base For Acrylic Plastics
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
DOD-D-1000	Drawing, Engineering and Associated List

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specification Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Bracing, Blocking and Waterproofing with Appropriate Test Methods

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

AIR FORCE-NAVY AERONAUTICAL

AN316	Nut, Jam, Hexagon

DRAWINGS

AIR FORCE

47A2330	Strap, Ground Astrodome
48A2184	Anchor, Strap Astrodome Grounding
49A22675	Decalcomania-Installation and Removal Date Accessories

* (Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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* AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-3951-82 Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.1.1 Standard parts. AN or MIL standard parts shall be used whenever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they develop suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Materials. Materials shall conform to the applicable specification and shall be as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.3.2 Treatment of plastic materials. Plastic materials used in the formation of the dome shall be preheated prior to forming for a period of not less than 30 minutes, at a temperature of not less than 300°F. The plastic material shall be formed immediately upon removal from the preheating chamber in order to avoid cooling of material. After placing the heated plastic material in the mold for forming of the dome, means shall be provided for preventing rapid cooling of the dome such as may be caused by air drafts. The dome shall remain in the forming fixture until the dome has reached room temperature. The dome shall be formed and cooled in a room whose temperature is not less than 60°F.

3.3.3 Forming fixture. The part of the forming fixture in contact with the heated plastic material shall be constructed of a material which is a poor heat conductor. Metal shall not be used.

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3.3.4 Protective treatment. When materials are used in the construction of the dome that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.4 Design. The dome, including the sextant suspension, shall be designed for use on nonpressurized aircraft, and shall conform to Figures 1 and 2.

3.5 Construction. The dome shall be constructed so that no parts will work loose in service. The dome shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 The dome shall be constructed so that adjustments and repairs can be easily made by personnel of operating units and overhaul bases.

3.5.2 The dome, exclusive of the metallic attachments which are herein after referred to as the sextant suspension, shall be free blown of plastic sheet conforming to MIL-P-5425, Type I.

3.5.3 Fabrication. The dome shall be fabricated in accordance with MIL-P-6997 and the following:

3.5.3.1 Optical symmentry. The dome shall be symmetrical about a vertical axis through its apex.

3.5.3.2 Radius of curvature. The spherical radius of curvature shall be approximately 11 inches. From point to point over the surface of the dome, the variation in radius from 11 inches shall not exceed $\pm$ 1/4 inch.

3.5.3.3 Dome surfaces. The curved surfaces of the dome shall have a high polish and shall be free from scratches, flaws, blemishes, or dull spots which might interfere with sextant observations taken through the dome.

3.5.3.4 Refractive index. The finished dome shall have a sensibly uniform index of refraction and shall give a clear definition of the image when observed through a transit or theodolite.

3.5.4 Sextant suspension. A suitable mounting as shown on Figure 2 shall be provided at the apex of the dome for attaching a sextant suspension arm in a horizontal position.

3.5.4.1 The mounting shall be such as to permit the sextant suspension arm to rotate about a vertical axis.

3.5.4.2 The mounting shall be made of brass or other substitute material suitable for the purpose. When nonmetallic substitutes are used in the construction of the mounting, the method of mounting shall be subject to the approval of the procuring agency.

3.5.5 Flange. The flange shall be an integral part of the dome and shall be constructed in accordance with Figure 1.

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3.5.5.1 The lower surface of the flange shall be flat and no part of the flange shall depart from the plane defined by the inside edge by more than 1/16 inch.

3.5.6 Conductor strip. A conductor strip constructed in accordance with Drawing No. 47A2330 shall be provided against the inside surface of each dome from its apex to its base. It shall be located in the vicinity of the calibration card. One end of the conductor strip shall be placed around the stud at the dome apex between metal plate and the plastic, the other end of the conductor strip shall hang free for use in grounding to aircraft.

3.5.6.1 Fastening conductor strip. The conductor strip shall be fastened in position by means of 3 plastic strips placed over the conductor strip. The plastic strips shall be constructed in accordance with Drawing No. 48A2184, and shall be so located as to permit alignment of the conductor strip with a bolt hole in the astrodome flange. The strips shall be equally spaced along the conductor strip and shall be permanently bonded to the dome by means of adhesive conforming to MIL-A-8576. The quality of adhesive used shall be such that the strength of the dome is not decreased to such an extent that the dome will not pass all of the tests specified herein.

3.5.7 Data and information card. A card shall be securely attached to the dome near its base. The card shall be protected by a sheet of transparent acrylic plastic bonded to the dome with adhesive in accordance with MIL-A-8576. The quantity of adhesive used shall be such that the strength of the dome is not decreased to such an extent that the dome will not pass all of the tests specified herein. The dimensions of the acrylic sheet shall not exceed 0.060 by 1-1/2 by 3 inches. The following information shall be permanently and legibly marked on the card.

3.5.7.1 The following information, unless otherwise specified, shall be printed in black on a white background on the side of card visible from inside of dome:

Calibration data showing the mean values of corrections to the nearest whole minute of the maximum and minimum vertical deviations observed for each 10 degree interval of altitude from 20 to 80 degrees as the dome is rotated in azimuth through 360 degrees. The sign used (+ or -) shall indicate the operation to be performed on the observed altitude to obtain the correct altitude. The following rule shall apply in determining the sign of correction indicated on calibration card: After insertion of the dome in line of sight, if the objective end of telescope must be raised to again effect collimation between crosshairs of the telescope and collimator image, then the altitude value obtained when sighting sextent through dome will be too great, hence the correction to be indicated on the calibration card is minus. This rule shall be valid when sights are taken through the small portion of the dome as indicated in Figure 2 and provided that the base of the dome is turned toward the telescope.

3.5.7.1.1 The following information shall be printed in red capital letters not less than 3/32 inch in height:

DO NOT SAND OR BUFF; WAX AND WASH ONLY
THIS DOME SUITABLE FOR USE IN NONPRESSURIZED
AIRCRAFT ONLY

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3.5.7.2 The following information shall be printed in red capital letters not less than 3/32 inches in height on the reverse side of card visible from outside of dome:

DO NOT SAND OR BUFF; WAX AND WASH ONLY

3.5.8 Installation and removal date. A card conforming to Drawing No. 49A22675 shall be permanently attached to the dome adjacent to the side of the date and information card. It shall be installed in a manner that the printed matter on the card may be read from within the dome, and that service personnel, when necessary, may add required information to the card.

3.6 Performance. The dome shall be capable of meeting the performance requirements of Section 4.

3.6.1 Optical deviations. For each altitude of the light beam as shown on Figure 3, the total variation of vertical deviation as the dome is rotated in aximuth shall not exceed 8 minutes of arc when the light beam is passed through the dome as specified in Section 4.

3.7 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Change in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.8 Screw threads. All screw threads shall be in accordance with MIL-S-7742.

3.9 Finishes and protective coating:

3.9.1 When aluminum alloy is used, it shall be treated in accordance with MIL-A-8625.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Nameplate.. A nameplate, permanently and legible filled with the following inforamtion shall be securely attached to the dome:

Dome, Navigator's Observing, Non-Pressurized Aircraft
Type A-3A
Specification MIL-D-5813B
Mfr's Part No._____
Mfr's Serial No._____
Order No._____
Stock No._____
(Mfr's Name or Trademark)
U. S. Property

3.11 Workmanship:

3.11.1 General. The dome, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of wiring, marking of parts and assemblies, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

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3.11.2 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the dome, they shall be held or limited accordingly.

3.11.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

3.11.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

* 3.12 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

* 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the domes shall be classified as acceptance tests.

4.3 Acceptance tests shall consist of individual tests.

4.3.1 Examination of product. Each dome shall be examined to determine compliance with this specification with respect to material, workmanship, dome surface, and dimensions.

4.3.2 Optical quality of the image. The quality of a circular collimator image or of a circle substituted for the grid shall be observed when sighting a transit or theodolite through the dome. It shall be possible to focus the transit and view the circle. If the circle cannot be seen or is distorted in shape, this shall be considered cause for rejection of the dome. The optical quality of the image shall be observed at the time the optical deviation test is being conducted.

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4.3.3 Optical characteristics determination. The optical characteristics of the dome shall be determined by sighting through a transit or theodolite at a collimator image as shown on Figure 4, or by use of a transit or theodolite and a lighted grid with spacings on the grid each equivalent to 1 minute of arc at the fixed distance the grid is located from the theodolite. Before the dome is inserted in the line of sight, the items shall be adjusted so that the crosshair of the theodolite and the collimator image, or zero reference on grid, appear to the superimposed. Provisions shall be made on dome support for rotating the dome in altitude through 90 degrees and continuously in azimuth through 360 degrees. It shall be possible to determine in minutes of arc the amount the reticle is removed from the collimator image when the dome is inserted in the line of sight. When this test is conducted, the variations in vertical deviation at any altitude shall not exceed 8 minutes of arc.

4.3.4 Optical deviations. Each dome shall be inserted in the line of sight as shown on Figure 4. The optical deviations for each dome shall be measured at each of the angles shown on Figure 3 $\pm$ 1 degree. The dome shall be continuously rotated 360 degrees in azimuth, observations being made at all positions of the dome. The mean value of the nearest whole minute of the maximum and minus vertical deviation determined at each altitude reading shall be entered on the calibration card as specified in 3.5.7.1. The total variation of vertical deviation as the line of sight is rotated in azimuth shall not exceed 8 minutes of arc.

4.4 Retest. Rejected domes shall not be resubmitted for inspection without furnishing full particulars concerning previous rejection and measures taken to overcome the defects.

* 4.5 Inspection of packaging. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified herein.

* 5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified in the contract or order each dome shall be preserved and packaged Method III, in accordance with MIL-P-116; in unit containers conforming to PPP-B-636, Type CF, Class - Domestic, Grade 275.

* 5.1.2 Industrial Packaging. Unless otherwise specified in the contract or order, each dome shall be preserved and packaged in accordance with the provisions established in ASTM-D-3951-82.

5.2 Packing. Packing shall be Level A, C or industrial, as specified (see 6.2).

5.2.1 Level A. Domes preserved and packaged as specified in 5.1.1. shall be packed in export type shipping containers conforming to PPP-B-601. As far as practicable, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitation of the specification. Closure and

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strapping shall be in accordance with the appendix of the specification. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level C. Domes preserved and packaged as specified in 5.1.1 shall not be overboxed for domestic shipment. The unit container, closed and strapped in accordance with the appendix to the container specification, shall be the shipping container.

* 5.2.3 Industrial. Packing will be in accordance with the provisions established in ASTM-D-3951-82.

5.3 Physical protection. Cushioning, blocking, bracing and bolting, shall be in accordance with MIL-STD-1186, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. Drop tests of MIL-STD-1186 shall be waived when preservation, packaging and packing of the item is for immediate use or when drop tests of MIL-P-116 are applicable.

* 5.4 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM-D-3951-82 as appropriate.

6. NOTES

6.1 Use. The Type A-3A dome covered by this specification is intended for use on non-pressurized aircraft to afford the navigator a means for observing celestial bodies for navigational purposes.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Subject term (key word) listing.

Dome, Navigator's Non pressurized aircraft
Dome, Non pressurized aircraft
Dome, Navigator's Type A-3A
Observing dome, Non pressurized aircraft
Type A-3A, Dome, Navigator's, non pressurized aircraft

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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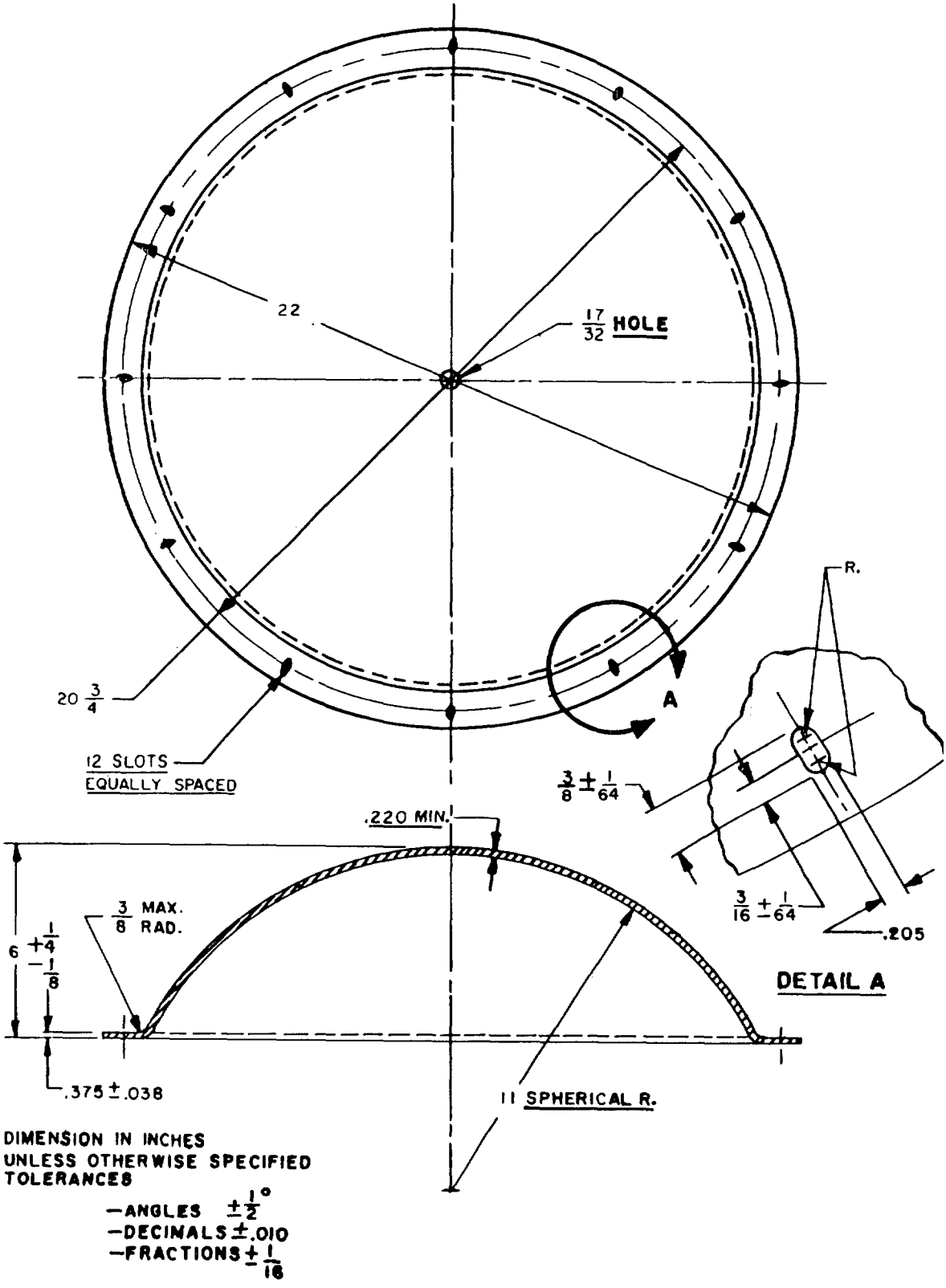
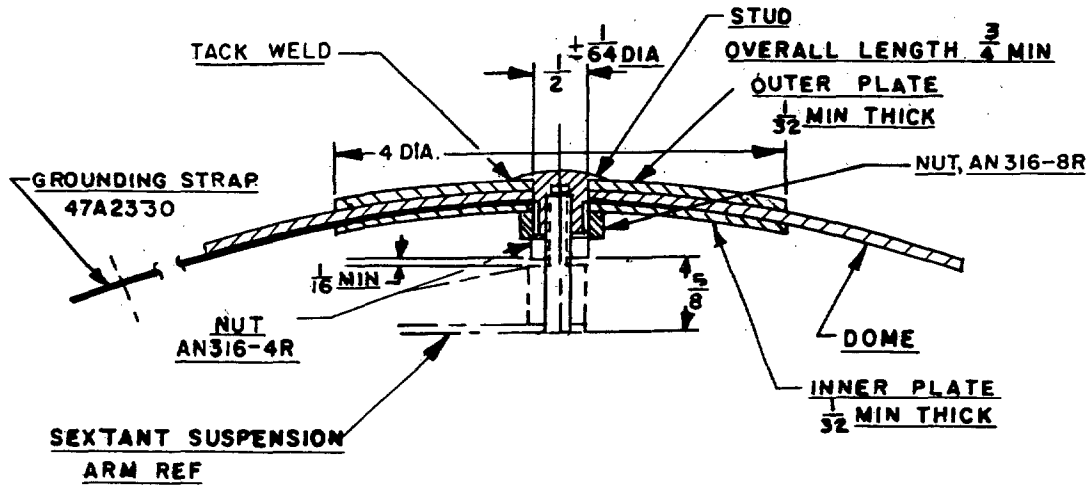


FIGURE 1. Dome outline.

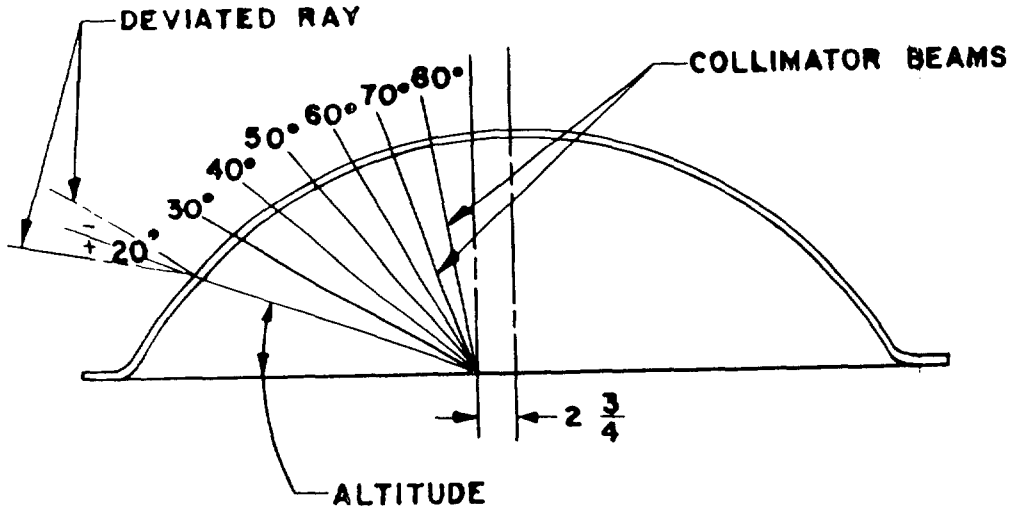
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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES: FRACTIONS ± 1/64

FIGURE 2. Sextant suspension.

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES:
—ANGLES— $\pm \frac{1}{2}°$
—DECIMALS—
—FRACTIONS— $\pm \frac{1}{64}$

FIGURE 3. Optical deviations calibration.

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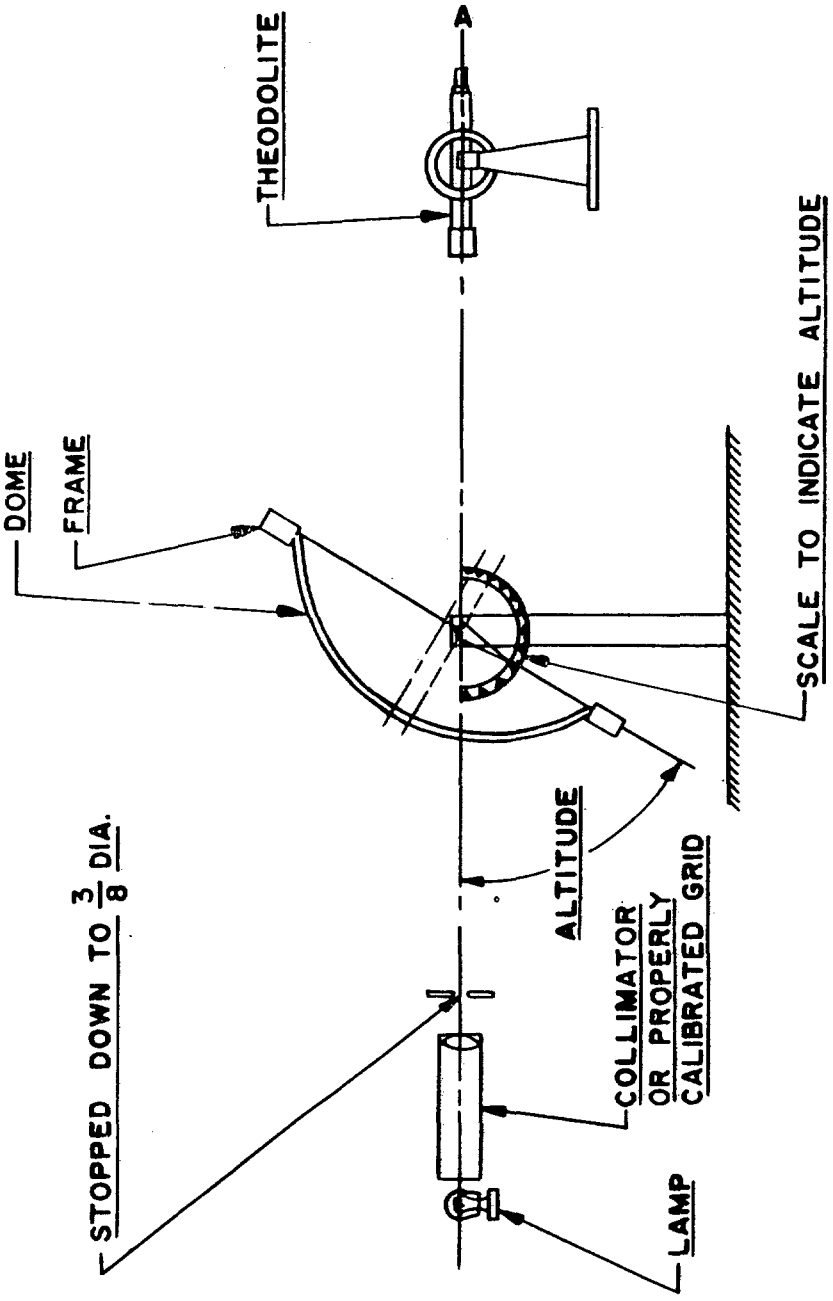


FIGURE 4. Test equipment.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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